Kieran Fraser, PhD

Researcher, Data Scientist & Engineer. Dublin, Ireland.

Summary

A researcher and engineer working across AI, security and human-computer interaction. I graduated from the University of Dublin, Trinity College with a PhD in Computer Science for research on synthetic data generation and push-notification technology (ETHOS Push). I then spent two years applying that research to commercial use as a Research Assistant at Trinity's ADAPT Centre, publishing jointly at leading Mobile, HCI and NLP conferences. From 2022 I was a Research Engineer at IBM on the Security and Privacy of AI team, working on adversarial robustness and red-teaming for vision and large-language models. Since November 2026 I've been a Staff Machine Learning Engineer at PayPal, working on onboarding and early-lifecycle fraud detection.

Education

PhD Computer Science — University of Dublin, Trinity College

SEP 2016 – MAR 2021 · DUBLIN

Thesis: *ETHOS Push* — EmpaTHetic Orchestration of Scrutable Push. A framework for the empathetic orchestration of scrutable push-notifications, motivated by a lack of empathy in the "push" communication space.

MAI Computer Engineering — University of Dublin, Trinity College

SEP 2015 – MAY 2016 · DUBLIN

Graduated with Distinction.

BAI Computer Engineering & BA Mathematics — University of Dublin, Trinity College

SEP 2011 – MAY 2015 · DUBLIN

Graduated with Second Class Honours, First Division (2.1).

Employment history

Staff Machine Learning Engineer, PayPal NOV 2026 – PRESENT · DUBLIN

Onboarding and early-lifecycle fraud detection. Mentored two interns, contributed to a portfolio redesign, filed two patent applications, and started a small patent group in the Dublin office.

Research Engineer, IBM AUG 2022 – NOV 2026 · DUBLIN

Research Engineer on the Security and Privacy of AI team at the Dublin Research Lab. Contributed to research, products, patents and open-source tools enhancing the adversarial robustness and fairness of vision and large-language models, plus new methods for synthetic data generation and agentic AI.

Research Assistant, ADAPT Centre for Digital Content Technology
SEP 2020 – AUG 2022 · DUBLIN

Awarded Enterprise Ireland funding for a Feasibility Study and subsequent Commercialisation project in Empathetic Push-notifications (EmPushy). Recruited a small team and led market research, trial-partner liaison, MVP development and analysis of 3M+ notifications, presenting findings to partner CTOs and R&D leads. Gained experience with AWS, GDPR compliance, IP documentation and project management.

Android Engineer, Lumafit JUN 2016 – AUG 2016 · DUBLIN

Summer internship migrating a local Android database to a real-time cloud backend, enabling an offline app experience and improving efficiency.

Executive Officer, Revenue Commissioners JAN 2015 – JUN 2015 · DUBLIN

Internship at the ICT and Logistics Division, overhauling internal web applications with the Spring Framework to improve customs department efficiency.

Data Analyst, Aon Centre for Innovation & Analytics MAY 2014 – JUL 2014 · DUBLIN

Summer internship building a Business Intelligence dashboard (SQL, Tableau), presented to ACIA's CEO, CBO and CCO to improve customer targeting and conversion.

Post-processing Engineer, Septentrio JUN 2013 – AUG 2013 · LEUVEN

Summer internship contributing to in-house tests and continuous integration (Jenkins) for satellite navigation software.

Skills

Machine Learning EXPERT

Data Analysis EXPERT

Synthetic Data Generation EXPERT

Cloud Computing EXPERT

Academic Writing EXPERT

Research EXPERT

Adversarial AI EXPERIENCED

Large-language Models EXPERIENCED

IT Project Management EXPERIENCED

Lean Startup EXPERIENCED

Technology

Python, Java, C++, JavaScript 10+ YEARS

HTML5, CSS, SCSS, Node, React, jQuery 10+ YEARS

SQL & NoSQL 10+ YEARS

Adobe Creative Suite 10+ YEARS

Docker 8 YEARS

Google Cloud Platform 3 YEARS

Amazon Web Services — IAM, Amplify, SageMaker, S3, DynamoDB, Cognito, Athena, Glue, Lambda, API Gateway, ECR, Batch, EC2

4 YEARS

Microsoft Azure 1 YEAR

Selected publications

Full list, including workshop papers and whitepapers, is on the Work page and Google Scholar.

Adversarial Prompt Evaluation: Systematic Benchmarking of Guardrails Against Prompt Input Attacks on LLMs

G. ZIZZO, G. CORNACCHIA, K. FRASER, ET AL. · 2025

Granite Guardian: Comprehensive LLM Safeguarding

I. PADHI, M. NAGIREDDY, G. CORNACCHIA, ET AL. · 2024/25

Attack Atlas: A Practitioner's Perspective on Challenges and Pitfalls in Red Teaming GenAI

A. RAWAT, S. SCHOEPF, G. ZIZZO, G. CORNACCHIA, K. FRASER, ET AL. · 2024

Extracting and Understanding Call-to-actions of Push-notifications

B. ESTEVES, K. FRASER, S. KULKARNI, O. CONLAN, V. RODRÍGUEZ-DONCEL · NLDB 2022

ETHOS Push: An Approach toward Empathetic Orchestration of Scrutable Push — PhD thesis, Trinity College Dublin

2021

Enticing Notification Text & the Impact on Engagement

K. FRASER, O. CONLAN · UBITTENTION 2020

Scrutable & Persuasive Push-Notifications

K. FRASER, B. YOUSUF, O. CONLAN · PERSUASIVE TECHNOLOGY 2019

A Context-Aware, Info-Bead and Fuzzy Inference Approach to Notification Management

K. FRASER, B. YOUSUF, O. CONLAN · UEMCON 2016 · BEST PAPER AWARD

Patents

Data quality assessment and transformation in privacy-preserving federated learning

IBM

Model emergent capability analysis IBM

Securing retrieval-augmented generation IBM

Utility-preserving anonymisation of visual content IBM

Two additional applications filed during my time at PayPal — not yet public. PAYPAL

Full inventor record on Google Patents INDEX

Hobbies

CrossFit, golf, tag rugby, art, photography, VR and photogrammetry — more on the Elsewhere page.

Links

LinkedIn

GitHub

Google Scholar

kieranfraser.ie